

COURSE 2B - DA-to-DE Bridge

# B04 Protected Practice Review

Open one review only after the matching genuine attempt is saved

## Fast-track purpose

This review diagnoses reasoning. It does not replace your work. After reading the matching review, close it and complete the changed retry without copying wording.

## Locked learning loop

Why -> picture -> new words -> watch -> analyst-to-engineer transition -> follow once -> observe -> explain -> mistakes -> recover -> fresh task -> validate -> teach back -> protected review -> changed retry -> evidence/route.

INTERNAL QA RC1 - not Course 2B FINAL

# Review B04-L01 - Operational vs analytical workloads: OLTP and OLAP

**Unlock condition:** saved attempt exists in practice sheet Workload\_Contract. If not, close this review now.

## What strong work demonstrates

Correctly separates event-recording workload from historical decision workload and names source-system risk.

## Common weak patterns

Calling every SQL database a warehouse; running dashboard-style full scans directly on the app database; treating OLTP/OLAP as vendor brands.

## Targeted repair

Ask: is this action recording/changing one business event now, or summarizing many events to make a decision? Then identify who is harmed if the query is slow.

## Changed retry - review closed

Fresh domain: clinic appointments, payments, weekly no-show analysis and annual capacity planning.

## Evidence check

Every row has workload type + reason + source-risk note. At least one analytical request has an explicit "do not run directly on operational source" boundary. Save the changed retry separately; do not overwrite the first attempt.

## Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

# Review B04-L02 - Warehouse, lake and lakehouse mental models

**Unlock condition:** saved attempt exists in practice sheet Storage\_Fit. If not, close this review now.

## What strong work demonstrates

Chooses storage pattern from data shape, consumers, governance, access and replay needs, with a defensible rejected alternative.

## Common weak patterns

Calling a folder of CSVs a governed lakehouse; assuming a lake is automatically cheaper overall; choosing a warehouse for raw binary/unstructured archives without a reason.

## Targeted repair

Write four questions: What data shape? Who queries it? How much governance/transaction reliability? Must raw history remain cheaply replayable?

## Changed retry - review closed

Fresh domain: e-commerce raw images/events, governed finance metrics and ML feature history.

## Evidence check

Every decision names at least three requirements and one trade-off; one rejected alternative is defended. Save the changed retry separately; do not overwrite the first attempt.

## Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

## Review B04-L03 - Business keys, surrogate keys and SCD history

**Unlock condition:** saved attempt exists in practice sheet History\_Keys. If not, close this review now.

### What strong work demonstrates

Separates business identity from warehouse version identity and prevents ambiguous historical intervals.

### Common weak patterns

Using a business key as if it uniquely identifies every historical version; Type 2 for every typo; overlapping effective intervals; silently deleting duplicate change rows.

### Targeted repair

Ask whether the old value must remain historically true. If yes, history likely matters. Then state the source identity separately from the warehouse version identity.

### Changed retry - review closed

Fresh domain: provider clinic change, spelling correction and specialty-history requirement.

### Evidence check

Business key stated; history choice justified per attribute; no ambiguous active interval; duplicate change trap is surfaced rather than hidden. Save the changed retry separately; do not overwrite the first attempt.

### Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

## Review B04-L04 - ETL vs ELT and transformation boundaries

**Unlock condition:** saved attempt exists in practice sheet Transform\_Boundaries. If not, close this review now.

### What strong work demonstrates

Names the durable landing point and transformation boundary and can explain what is replayable after a bad rule.

### Common weak patterns

Saying ELT means "no cleaning"; calling every sequence ETL; transforming raw evidence in place; assuming cloud always means ELT.

### Targeted repair

Mark the first durable copy. Then ask: if tomorrow's rule changes, what can I replay without requesting the source again?

### Changed retry - review closed

Fresh domain: subscription billing with raw landing and a sensitive-field pre-load rule.

### Evidence check

Flow has explicit source, durable point, transform boundary, curated output and replay statement. Save the changed retry separately; do not overwrite the first attempt.

### Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

## Review B04-L05 - Layered analytical systems: raw, staged, conformed and serving

**Unlock condition:** saved attempt exists in practice sheet Layer\_Contracts. If not, close this review now.

### What strong work demonstrates

Defines layer contracts by allowed state, transformations, consumers and promotion quality gates rather than color labels alone.

### Common weak patterns

Editing raw source rows; putting business KPIs into raw; using layer names with no consumer/quality contract; assuming every project needs exactly three physical layers.

### Targeted repair

For each layer answer: what can enter, what may change, who trusts it, and what blocks promotion?

### Changed retry - review closed

Fresh domain: logistics source events -> normalized shipments -> governed SLA mart.

### Evidence check

Every layer has allowed state, transformation boundary, consumer, quality gate and recoverability statement. Save the changed retry separately; do not overwrite the first attempt.

### Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

## Review B04-L06 - Batch, micro-batch and streaming trade-offs

**Unlock condition:** saved attempt exists in practice sheet Latency\_Plan. If not, close this review now.

### What strong work demonstrates

Starts from freshness/recovery SLA and chooses batch/micro-batch/streaming with late-data and replay reasoning.

### Common weak patterns

Choosing streaming because it sounds advanced; confusing refresh frequency with source event time; ignoring late data; promising zero latency.

### Targeted repair

Start with "How stale can this safely be?" Then ask what happens if processing stops for 20 minutes.

### Changed retry - review closed

Fresh domain: hospital telemetry alert, hourly utilization and monthly finance close.

### Evidence check

Each mode has SLA + late-data + replay + cost/complexity rationale. Save the changed retry separately; do not overwrite the first attempt.

### Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

# Review B04-L07 - Lineage, metadata, ownership and observability basics

**Unlock condition:** saved attempt exists in practice sheet Lineage\_Ownership. If not, close this review now.

## What strong work demonstrates

Provides traceable upstream/downstream lineage, run metadata, ownership and failure-impact routing.

## Common weak patterns

Treating a folder path as complete lineage; collecting logs with no run ID; naming an owner with no responsibility boundary; tracking only failures and not normal row/freshness evidence.

## Targeted repair

Start from the wrong output and walk upstream one transformation at a time. Write the minimum evidence you would need at each hop.

## Changed retry - review closed

Fresh domain: support tickets -> clean events -> SLA metrics -> executive dashboard.

## Evidence check

Chain has upstream/downstream links, owner, freshness expectation, run ID evidence and one failure-impact route. Save the changed retry separately; do not overwrite the first attempt.

## Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

# Review B04-L08 - Data-quality contracts: block, warn and reconcile

**Unlock condition:** saved attempt exists in practice sheet Quality\_Contract. If not, close this review now.

## What strong work demonstrates

Connects each quality rule to severity, action, owner and evidence; reconciliation is independent and issues are not hidden.

## Common weak patterns

Replacing errors with zero; deleting orphans to make counts match; marking every issue BLOCK; using one grand total as proof of all correctness.

## Targeted repair

For each defect ask: can a consumer make a materially wrong decision if this passes? If yes, block or isolate until explicitly resolved.

## Changed retry - review closed

Fresh domain: marketplace duplicate seller history + orphan payout + GMV reconciliation.

## Evidence check

At least one BLOCK, one WARN/QUARANTINE, one independent reconciliation and named owner. No issue is hidden by deletion. Save the changed retry separately; do not overwrite the first attempt.

## Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.

# Review B04-L09 - Replay, backfill and warehouse failure-recovery reasoning

**Unlock condition:** saved attempt exists in practice sheet Recovery\_Runbook. If not, close this review now.

## What strong work demonstrates

Preserves raw evidence, bounds recovery scope, protects state, prevents duplicates and proves post-recovery correctness.

## Common weak patterns

Deleting bad output before preserving evidence; advancing checkpoint before reconciliation; rerunning unbounded history; validating only the metric that originally failed.

## Targeted repair

Write seven verbs in order: detect, contain, identify scope, recover from preserved input, validate independently, promote, document.

## Changed retry - review closed

Fresh domain: two missing support-event days where downstream daily aggregates already exist.

## Evidence check

Runbook names affected window, source evidence, state/checkpoint handling, duplicate prevention, at least two independent controls and promotion/rollback rule. Save the changed retry separately; do not overwrite the first attempt.

## Mastery decision

PASS only when the changed retry is independently correct, validated and explainable. A weak first attempt remains in history even after repair.